

Mahler Measure and Dynamical Systems

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1 Introduction

In this paper, we introduce an operator, denoted as Ξ , which acts on polynomials by exponentiating their roots, allowing us to study the Mahler measure from a new perspective. We focus on monic polynomials $P(z) = \prod_i (z - r_i)$, where r_i are the roots of P . In many ways the logarithmic Mahler measure is 'more natural', and so most of our focus will go specifically there. A nice result we will present is that the logarithmic Mahler measure of the root-exponentiated polynomial $\Xi(P)$ can be expressed as a sum of the positive real parts of the roots of P , specifically,

$$m(\Xi(P)) = \sum_{\Re(r_i) > 0} \Re(r_i).$$

This relationship highlights the significance of Ξ , and why it is a very natural way to later use the Mahler measure to quantify the stability of a system. We conclude with a discussion on potential areas for future research that may arise from the framework established in this paper, emphasizing the broader implications of our findings in the study of dynamical systems and their stability, like potential geometric interpretations of stability.

We reformulate Lehmer's conjecture into asking what the smallest τ is such that

$$\ln \text{Vol} \mathcal{P}(\Omega_u(A)t) \geq t \ln \tau$$

1.1 Root-Exponentiating Operator

Let $P \in \mathbb{C}[z]$ be a complex polynomial with roots r , then

$$P(z) = \sum_{i=0}^n a_i z^i = \prod_{i=0}^n (z - r_i) \tag{1}$$

throughout this paper we will consider only monic polynomials. Now we will consider a new operator Ξ that exponentiates the roots of P ,

Definition 1.

$$\Xi(P) = \prod_i (z - e^{r_i}) \tag{2}$$

The operator Ξ is multiplicative when acting on products of polynomials such that

$$\Xi(PQ) = \Xi(P)\Xi(Q) \tag{3}$$

This can easily be seen as a consequence of the fact that a product of polynomials will contain all the roots of its constituent polynomial factors.

Lemma 1.1. *We know that the exponential function e^x is defined for all $x \in \mathbb{R}$ and can be extended to $x \in \mathbb{C}$. Hence $\Xi(P)$ is well-defined for every complex polynomial.*

Lemma 1.2. *There exists an inverse of Ξ as long as $\ln(r)$ exists for all roots of the polynomial it acts on. We can see that for $\Xi^{-1}(\Xi(P)) = \Xi(\Xi^{-1}(P)) = P$ to hold it must be*

$$\Xi^{-1}(P) = \prod_i (z - \ln(r_i)) \quad (4)$$

Lemma 1.3. *If P is a reciprocal polynomial, that is, it satisfies the identity $P(z) = z^{\deg(P)} P(z^{-1})$, then it is not a reciprocal polynomial under the action of Ξ .*

Proof. First we need to note that $\Xi(x) = x - 1$. To prove that the reciprocal property of some polynomials is not conserved under the action of Ξ we use the multiplicative property of Ξ and apply it to both sides to see that

$$\Xi(P(z)) = \Xi(z^{\deg(P)}) \Xi(P(z^{-1})) = (z - 1)^{\deg(P)} \Xi(P(z^{-1})) \quad (5)$$

which we can immediately see is not a reciprocal polynomial as it does not satisfy the requirement. This results in the fact that every P that satisfies

$$P(z) = (z - 1)^{\deg(P)} P(z^{-1}) \quad (6)$$

is reciprocal under the action of Ξ^{-1} , as long as $\ln(r)$ exists for all roots. $\square$

Lemma 1.4. *We can further prove by looking at Lemma 1.3 that there exist no reciprocal polynomial of the form $\Xi(P)$*

Proof. Assume that it is true, then by Lemma 1.3 (5) would require some polynomial $q(z) = z^m$ to exist such that $\Xi^{-1}(q)$ exists. This is a contradiction as it requires $\Xi^{-1}(q) = (z - \ln(0))^m$ to be well-defined, and hence cannot exist. $\square$

1.2 Mahler Measure

Let M denote the Mahler measure of a polynomial defined as

$$M(P) = \exp \left(\int_0^1 \ln |P(e^{2\pi i \theta})| d\theta \right) = |a_n| \prod_i \max(1, |r_i|) = \prod_i \max(1, |r_i|) \quad (\text{For monic } P) \quad (7)$$

Then the logarithmic Mahler measure is given as $m(P) = \ln M(P)$. The Mahler measure is multiplicative so we can see that $M(\Xi(PQ)) = M(\Xi(P))M(\Xi(Q))$ and $m(\Xi(PQ)) = m(\Xi(P)) + m(\Xi(Q))$. This is useful to know, and makes the computation of certain polynomials whose factors are known easier to compute and study algebraically.

Theorem 1.5. *The logarithmic Mahler measure of $\Xi(P)$ is given by*

$$m(\Xi(P)) = \sum_{\Re(r) > 0} \Re(r_i) \quad (8)$$

Proof. To derive this we simply see from (7) that

$$m(\Xi(P)) = \ln \prod_i \max(1, |e^{r_i}|) \quad (9)$$

We can see that $|e^r| = e^{\Re(r)}$, since e^r is always positive for any real r . For complex roots with imaginary part $|\Im(r)| > 0$ we need to use the identity, $e^r = e^{\Re(r)} e^{i\Im(r)}$ to see that $|e^r| = |e^{\Re(r)}| |e^{i\Im(r)}| = e^{\Re(r)}$. Therefore, one can see that

$$m(\Xi(P)) = \ln \prod_i \max(1, e^{\Re(r_i)}) = \ln \prod_i e^{\max(0, \Re(r_i))} = \sum_{\Re(r_i) > 0} \Re(r_i) \quad (10)$$

This concludes the proof $\square$

Corollary 1.5.1. *We have the strict inequality,*

$$\max(0, -a_{n-1}) \leq m(\Xi(P)) \leq \sum_i |r_i| \quad (11)$$

where a_{n-1} is the $(n-1)$ -th coefficient of P .

Proof. This follows from (8) and Vieta's formula for the sum of roots for a monic P . $\square$

2 Dynamical Systems

We will concern ourselves with the homogeneous linear dynamical systems of the form

$$\dot{x}(t) = Ax(t) \quad (12)$$

due to the power this form has as an approximation for other systems. x is the vector/function representing the state of the system at time t , and A is a constant matrix of the system. A higher-order homogeneous linear differential equation can be expressed in the matrix form $\dot{\mathbf{x}} = A\mathbf{x}$ by introducing new variables to represent the derivatives. This transformation is particularly useful for analyzing the system's behavior. Consider an n -th order homogeneous linear differential equation:

$$A_n x^{(n)} + A_{n-1} x^{(n-1)} + \dots + A_1 \dot{x} + A_0 x = 0,$$

where $x^{(k)}$ denotes the k -th derivative of $x(t)$, and $A_n, A_{n-1}, \dots, A_0$ are constants. Define new variables $x_1, x_2, \dots, x_n$ such that:

$$\begin{aligned} x_1 &= x, \\ x_2 &= \dot{x}, \\ x_3 &= \ddot{x}, \\ &\vdots \\ x_n &= x^{(n-1)}. \end{aligned}$$

The derivatives of these variables are:

$$\begin{aligned}\dot{x}_1 &= x_2, \\ \dot{x}_2 &= x_3, \\ &\vdots \\ \dot{x}_{n-1} &= x_n, \\ \dot{x}_n &= -\frac{1}{A_n} (A_{n-1}x_n + \cdots + A_1x_2 + A_0x_1).\end{aligned}$$

‘ The system can now be written in the form $\dot{\mathbf{x}} = A\mathbf{x}$, where:

$$\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix}, \quad A = \begin{bmatrix} 0 & 1 & 0 & \cdots & 0 \\ 0 & 0 & 1 & \cdots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \cdots & 1 \\ -\frac{A_0}{A_n} & -\frac{A_1}{A_n} & -\frac{A_2}{A_n} & \cdots & -\frac{A_{n-1}}{A_n} \end{bmatrix}.$$

3 Example

Consider the second-order differential equation:

$$\ddot{x} + 3\dot{x} + 2x = 0.$$

Define $x_1 = x$ and $x_2 = \dot{x}$. The system becomes:

$$\begin{aligned}\dot{x}_1 &= x_2, \\ \dot{x}_2 &= -2x_1 - 3x_2.\end{aligned}$$

In matrix form:

$$\dot{\mathbf{x}} = A\mathbf{x}, \quad \text{where} \quad A = \begin{bmatrix} 0 & 1 \\ -2 & -3 \end{bmatrix}.$$

4 Conclusion

This transformation allows higher-order differential equations to be analyzed using linear algebra techniques, with the matrix A encoding the system’s dynamics.

Proof. (Not a proof, but a simple example). To demonstrate the importance of linear systems of this form, without going into any details, consider perhaps one of the most famous filters; the Kalman filter,

$$\dot{\hat{x}} = A\hat{x} + Bu + K(y - C\hat{x} - Du)$$

Defining the error as $\varepsilon = x - \hat{x}$ (i.e the difference between filter state and actual state), and thus its derivative as $\dot{\varepsilon} = \dot{x} - \dot{\hat{x}}$. Substituting in the expressions for $\dot{x}$, $\dot{\hat{x}}$ and y yields,

$$\dot{\varepsilon} = (Ax + Bu) - (A\hat{x} + Bu + K((Cx + Du) - C\hat{x} - Du)) = (A - KC)\varepsilon$$

which, if we let $(A - K C) = M$ we get

$$\dot{\varepsilon} = M\varepsilon$$

This is just an example to illustrate its prevalence. Another would be that one can use this form of equation to approximate some non-linear dynamical systems. $\square$

We let $p_A(\lambda)$ denote the characteristic polynomial of A with roots being the eigenvalues λ_i .

Lemma 4.1. *All characteristic polynomials are monic.*

Proof. This can be easily seen from the definition

$$p_A(\lambda) = \det(\lambda I - A) = \sum_{\sigma \in \Sigma_n} \text{sgn}(\sigma) \prod_{i=1}^n (\lambda \delta_{i\sigma(i)} - A_{i\sigma(i)}) \quad (13)$$

showing that the leading coefficient of λ is the identity permutation and hence always one, and therefore p_A is always monic. $\square$

Theorem 4.2. *If A is diagonalizable, then we have the symmetry*

$$\Xi(p_A) = p_{e^A}(\lambda) \quad (14)$$

Proof. This is an immediate consequence of the fact that $e^A = X e^D X^{-1}$, as Ξ exponentiates the roots $\lambda \in D = \text{diag}(\lambda_1, \lambda_2, \dots, \lambda_i)$, and the fact that $e^D = \text{diag}(e^{\lambda_1}, e^{\lambda_2}, \dots, e^{\lambda_i})$, hence we can see $\Xi(p_A)$ as the characteristic polynomial of e^A . $\square$

Corollary 4.2.1. *The system (12) is stable iff all eigenvalues of A have a negative real part, and the system is unstable if some eigenvalue has a positive real part. This can be seen by the solution,*

$$x(t) = x_0 e^{At} \quad (15)$$

and if A is diagonalizable then

$$x(t) = \sum_i (v_i^T x_0) e^{\lambda_i t} v_i \quad (16)$$

where we can see better that the eigenvalue's real part determines the exponential scaling of the system in the direction associated to its eigenvector, which shows that if the system has an eigenvalue with positive real part, no matter how small, the system is asymptotically unstable.

Definition 2. *The stable manifold W^s is the set of all initial conditions x_0 for which $x(t)$ approaches the equilibrium point as asymptotically*

$$W^s = \left\{ x_0 \in \mathbb{R}^n : \lim_{t \rightarrow \infty} x(t) = 0 \right\}$$

This manifold is spanned by the eigenvectors v_i corresponding to $\Re(\lambda_i) < 0$.

Definition 3. *The unstable manifold W^u is the set of all initial conditions x_0 for which the solution $x(t)$ diverges from the equilibrium point asymptotically*

$$W^u = \left\{ x_0 \in \mathbb{R}^n : \lim_{t \rightarrow -\infty} x(t) = 0 \right\}$$

This manifold is spanned by the eigenvectors v_i corresponding to $\Re(\lambda_i) > 0$.

5 Instability

5.1 A Scalar Measure of Instability

It is easy to see that,

$$m(\Xi(p_A)) = m(p_{e^A}) = \sum_{\Re(\lambda) > 0} \Re(\lambda_i) \quad (17)$$

is a measure of the 'amount of instability' of the system associated to A , as it is simply a sum of the eigenvalues associated with instability. This provides an interesting scalar quantity for every system to study, however we can study further...

5.2 Geometric Interpretation of Instability

Next, consider associating an n -dimensional parallelepiped $\mathcal{P}$ with some matrix A , such that

$$\mathcal{P}(A) := \left\{ \sum_{i=1}^n a_i u_i : a \in [0, 1] \right\}$$

where u is such that $A = [u_1 \ u_2 \ \dots \ u_n]$ i.e the columns of A . Next, consider restricting the dynamics of the system (12) on W^u which would yield

$$\dot{y}(t) = \Omega_u(A)y(t) \quad (18)$$

where $\Omega_u(A)$ will denote the matrix of the restricted system containing all eigenvalues and eigenvectors that span W^u (similarly Ω_s would denote the stable analog). The solution to this system, like for (12) is given as

$$y(t) = y_0 e^{\Omega_u(A)t} \quad (19)$$

It is well-known that the signed volume of a parallelepiped defined by the columns of some matrix A is given by the determinant of A .

Now comes the interesting part. Consider the volume of the parallelepiped defined by the constant matrix of the restricted system (writing Ω_u for simplicity), specifically

$$\text{vol}(\mathcal{P}(\Omega_u t)) = \det(e^{\Omega_u t}) \quad (20)$$

by Jacobi's identity this gives us

$$\text{vol}(\mathcal{P}(\Omega_u t)) = \det(e^{\Omega_u t}) = \exp(\text{Tr}(\Omega_u t)) = \exp\left(\sum_{\lambda \in \Omega_u(A)} \lambda_i t\right) \quad (21)$$

notice,

$$\sum_{\lambda \in \Omega_u(A)} \lambda_i = \sum_{\substack{\Re(\lambda) > 0 \\ \lambda \in A}} \Re(\lambda_i) = m(\Xi(p_A)) \quad (22)$$

and therefore we have, fascinatingly, that

$$\text{vol}(\mathcal{P}(\Omega_u t)) = \exp(m(p_{e^A})t) = M(p_{e^A})^t \quad (23)$$

The volume of a parallelepiped associated with the unstable manifold of a linear dynamical system is given by the Mahler measure of the characteristic polynomial of the exponential of the original constant matrix of the system/the root-exponentiated characteristic polynomial of the original constant matrix. Furthermore, this states that the rate of growth of the volume of $\mathcal{P}$ associated to a neighborhood of W^u over time is given by the logarithmic Mahler measure as

$$\text{vol}(\mathcal{P}(\Omega_u t)) = \exp(m(p_{e^A})t) \quad (24)$$

6 Conjectured Connections

6.1 Entropy

Immediately, if we look at the logarithm of the volume, we get

$$\ln \text{vol}(\mathcal{P}(\Omega_u)) = m(p_{e^A}) \quad (25)$$

the logarithm of the volume of certain mathematical objects is closely related to entropy in some cases. Strangely, $m(P)$ is known to be related intimately with topological entropy. Furthermore, what's very interconnected with entropy in statistical mechanics is the partition function Z of a system. Seemingly unrelated, if we take a look at Dimer coverings, taking G to be a finite balanced bipartite planar graph, with edge weights μ_e for each edge $e \in G$. A Kasteleyn matrix $\mathcal{K}_G$ is a weighted, signed adjacency matrix of G , with horizontal edges weighted 1 and vertical edges weighted $\sqrt{-1}$, then the number of dimer coverings of G is given by the partition function

$$Z(G) = |\det(\mathcal{K}_G)| \quad (26)$$

specifically if G is biperiodic Theorem 3.5 in [1] says that

$$\ln Z(G) = m(p(z, w)) \quad (27)$$

basically saying that the growth rate of any toroidal dimer model is given by the Mahler measure of its characteristic polynomial p . This is extremely similar in form to (25), and in [1] it is detailed as the limit of entropy analogous to the partition function in statistical mechanics. Is there an equivalence? Furthermore, it is well known that m is related to volumes of polytopes in hyperbolic space. Further connections to entropy regarding automorphisms has also been established in [2]

6.2 Lehmer's Conjecture

Conjecture 6.1. *Given a polynomial $q \in \mathbb{C}$ such that $\Xi(q) = Q(x)$ Lehmer's conjecture then states that, given any dynamical system of the form*

$$\dot{x} = Ax$$

where $p_A(\lambda) = q(\lambda)$ so that $\Xi(p_A) = p_{e^A}(\lambda) = Q(\lambda)$ restricting the dynamical system to W^u , the conjecture states that there exists a $\tau > 1$ such that

$$\ln \text{vol}(\mathcal{P}(\Omega_u t)) \geq t \ln \tau \quad (28)$$

for all A .

6.2.1 Lehmer's Polynomial

In the case of the smallest known Salem number, that would give a polynomial $l \in \mathbb{C}$ such that

$$\Xi(l) = L(x) = x^{10} + x^9 - x^7 - x^6 - x^5 - x^4 - x^3 + x + 1 \quad (29)$$

i.e. Lehmer's polynomial. Then, given any dynamical system of the form $\dot{x} = Ax$ such that $\Xi(p_A) = p_{e^A}(\lambda) = L(\lambda)$ we get $\tau = 1.176280818\dots$

$$\text{vol}(\mathcal{P}(\Omega_u t)) \approx 1.176280818^t \quad (30)$$

Stating that, if $\tau = 1.176280818\dots$ would indeed be the smallest, then the rate of growth of a $\mathcal{P}$ associated to W^u is bounded by the smallest known Salem number.

It makes sense that there would exist a bound $\tau > 1$, as there would be no unstable manifold W^u if $\tau \geq 1$ (since if $\tau = 1$ such a system has no unstable components as the instability/volume does not grow), but why $\tau = 1.176280818\dots$??? It seems so much 'bigger' than one. Why not $\tau \geq 1 + \epsilon$ where $0 < \epsilon \ll 1$?

7 Questions

- Is there an equivalence between growth rate/instability/entropy of finite bipartite planar graphs and parallelepipeds as detailed in 3.2?
- Why is it that the Mahler measure's connections is regarding multivariable polynomials, however, for instability in dynamical systems (if there were an equivalence of the concepts) that the characteristic polynomial is in one variable? Can the reduction of a multivariable characteristic polynomial of one geometric object therefore be proven equivalent to another in one variable?
- Why, given any unstable dynamical system, does the instability/growth-rate of $\text{Vol}(\mathcal{P}(\Omega_u))$ need to be sufficiently larger than one (i.e. at least $0.176280818\dots$ larger)?
- What further connections can be established?

References

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